An Analysis of Collegiate Athletics Success on Admissions at the Division III Level

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Abstract

This paper aims to examine the relationship between athletic success in Division III programs and their respective admissions outcomes. I focus on the yield rate, defined as the percentage of admitted students who elect to enroll, as well as the number of applicants as response variables. Using a longitudinal panel dataset consisting of 425 Division III schools and 8 sports observed over a 10 year period (2013-2022), I aim to determine whether qualifying for the end-of-season NCAA tournament is associated with changes in the yield rate or number of applicants in the following year's admissions cycle, while controlling for various school-level characteristics such as tuition, percentage of students receiving financial aid, 75th percentile of SAT scores, and more. Using a Differences-in-Differences model incorporating two-way fixed effects, I do not find statistically significant evidence of such an association. There seems to be a slightly positive association with yield, and a slightly negative association with applications, though the magnitude is close to zero. Robustness checks, including regressions differentiating by sport and gender, as well as modern DiD models, corroborate these findings.

1 Introduction

The association between athletics success and academic outcomes at the collegiate level has long been fascinating to researchers. The correlation, also known as the Flutie Effect, aims to determine whether athletic success has a causal impact on various academic benchmarks. This has been widely studied in major Division I programs, which have greater media visibility and reach across the country. Originally, literature found that there were positive shocks in admissions trends following notable athletic successes, though recent studies have found dampened and smaller effects. However, the effect has been less studied in Division III schools, where visibility is reduced and athletes do not have access to athletic scholarships. This paper aims to determine whether there is a similar effect of athletic success on academic outcomes.

The yield rate, or the proportion of admitted students who end up accepting the offer of admission, is an important metric for higher education institutions. Higher yield signifies greater prestige, reputation, and enhances a school's perception, where incoming applicants perceive the institution as potentially more "desirable" than peer schools. All of these indicators can help a school improve their diversity and demographics of their first-year cohort. In addition, the number of applicants is another metric for schools to consider. This directly measures students' levels of interest in the school, and is not contingent on a student gaining admission to the school.

Unlike Division I schools, Division III schools do not offer athletic scholarships, and thus rely more on their institutional values and reputation to attract higher quality applicants. Though students at Division III schools tend to focus less on athletics and sporting events compared to Division I schools, determining whether there is an effect can help inform school decision-making and policies to enhance the student experience.

Section 2 introduces some background information to the Flutie Effect phenomenon, as well as numerous pieces of literature that initially corroborated the shock effect, though more modern research has questioned those findings. Section 3 lays out the sources of data used to compile school-level information during the time period. Section 4 introduces the methodology, while section 5 goes over the key findings and results. Finally, section 6 discusses long-term policy implications and possible improvements.

2 Background and Literature Review

2.1 Background

Analysis of the relationship between athletics success and institutional metrics dates back to 1984, when Boston College quarterback Doug Flutie threw a Hail Mary to beat rival Miami in a highly publicized college football game. Subsequently, Boston College saw a surge in applications, increasing by 30% in the following two years (Forbes). Subsequently, researchers became fascinated by the possibility that collegiate athletics programs succeeding could be causally linked with improvements in institutional variables, such as increases in applications, diversity, and cognitive levels. Research in the late 1900s and early 2000s seemed to corroborate this sentiment.

One of the first notable findings came from Murphy and Trandel (1994), where they analyzed the relationship between a Division I school's football record and the size and quality of the university's applicant pool. A higher winning percentage and more appearances in the postseason was statistically significantly correlated with increases in both the quantity and quality of the school's applicants in the following admissions cycle, indicating that athletics success can enhance a school's visibility and prestige. Toma and Cross (1998) also found similar positive shocks, determining that universities that field championship-winning teams experience increases in applications the following year. This was especially true for schools that did not experience much success in the preceding periods. Finally, Pope and Pope (2009) found a statistically significant increase in applications, but less of an effect on student quality. However, as new econometric methods have been introduced in recent times, these findings have come into question.

2.2 Recent Literature

Despite an abundance of research finding statistically significant increases in applications and other institutional outcomes due to athletics success, modern research has found negligible changes in admissions following athletic success, and in some cases, even decreases. Caudill et. al (2018) found that college football success results in inconsistent effects on applications, and a lack of statistically significant evidence of improvements in student quality. In addition, Hickman and Meyer (2016) found that athletics success did not improve retention rates at the university. While admissions outcomes may improve, student outcomes already enrolled at the college did not in a statistically significant way. Perhaps most alarming of all, Eggers et al. (2021) found that applications and enrollment increase as a byproduct of athletics success, but the academic quality of students decreases. For example, fewer students who ranked in the top 10% and top 25% in their respective high schools ended up enrolling. Although more students enrolled, the findings indicate a shift towards students with inferior academic backgrounds, that simply enjoy/prioritize sports more. Furthermore, the authors found that the teams that lost in a high-profile game also corresponded to increases in applications and enrollment. While "high-profile games" typically involve two very successful and talented programs, this raises the question whether winning is the true driver behind admissions increases, rather than simply being in the spotlight and visible to a wide audience, regardless of outcome.

Although there has been much discussion on the Flutie Effect, almost all research has been conducted at the Division I level, where athletics are more important, prominent, and visible to others. Little exploratory analysis has been done at the Division III level. A study by the Urban Institute in 2023 found that increased investments in athletics are negligibly correlated with enrollment changes at such schools, directly countering the effect at the D1 level. In addition, a study by McCannon (2025) found that starting a Division III football program had no effect on university enrollment. Despite football being the most popular sport in America, and especially at the D1 level, the same cannot be said for D3. I aim to contribute to existing literature by analyzing whether athletics success is associated with choosing to apply to a school, but also choosing to attend a school if admitted.

3 Data

3.1 Sources

The data for this analysis was compiled from a variety of sources. First, data was retrieved from the NCAA website containing a list of teams that qualified for the NCAA tournament in their respective sport. The sports are men's and women's basketball, soccer, lacrosse, and baseball and softball. This was to ensure an equal mix of men's and women's sports, while also focusing on the sports that are typically the most popular and visible to students. The teams were divided into two categories: Automatic Qualifiers (AQs) and at-large bids. Teams that won their conference automatically qualified to the NCAA tournament. However, to reward teams that perform well, but didn't win their conference, such teams receive an at-large bid, and can also participate in the NCAA tournament. Though the number of teams that qualify differ by sport, typically between 2/3 and 3/4 of the tournament field is comprised of automatic qualifiers. Data ranged from 2013 to 2022. Though there is NCAA qualification data from the 2023-2024 and 2024-2025 academic years, I did not have the corresponding IPEDS data to reflect admissions changes, and therefore did not include them in my study.

Institutional characteristics were aggregated from the Integrated Postsecondary Education Data System (IPEDS), founded by the Department of Education's National Center for Education Statistics. It contains data for many school-level characteristics. I focused on tuition (inflation-adjusted), percentage of students receiving financial aid, retention rate, student-to-faculty ratio, and the 75th SAT percentile. These covariates help explain differences between schools, sports, and gender. Data ranged from 2015 to 2024, to reflect the 2 year lagging of athletics success.

Data was also compiled from the Department of Education's Equity in Athletics Data Analysis (EADA) database, containing information about the expenditures for each athletic program at each school. It provides insights into which programs are justifying investment, and conversely which programs may require changes in spending. When combined with indicators of program success, it may inform institutions on whether they should reallocate resources to teams that more directly drive increases in admissions.

Finally, data was withdrawn from NCSA College Recruiting's list of current Division III schools.

Data was merged into one comprehensive dataset by their school name. The dataset was filtered to include only current Division III teams, excluding those that transitioned to another division or closed down, to maintain a balanced panel dataset. Further discussion of this step is mentioned in the limitations section.

Table 1 indicates descriptive statistics for the primary institutional covariates.

Table 1: Summary Statistics for Admissions

	(1)			
	mean	sd	min	max
Yield Rate	0.23	0.12	0.02	1.00
Acceptance Rate	0.67	0.22	0.03	1.00
Applicants	5626.99	7304.64	74.00	113578.00
Adjusted Tuition	50785.43	13316.00	2549.54	81531.00
Retention Rate	78.93	9.72	34.00	100.00
Student-Faculty Ratio	12.17	3.26	3.00	47.00
% of students receiving financial aid	82.15	16.69	0.00	100.00
75th SAT Percentile	1266.58	138.35	890.00	1600.00
N	4304			

On average, the yield rate is around 23%, while the average acceptance rate is around 67%. Schools retain approximately 79% of their students after their first year, with an average 2019-adjusted tuition of $50,785. Around 82% of first-year undergraduates receive some form of financial aid, and the average 75th SAT percentile for the incoming class is 1266. A college receives around 5600 applications on average, and has 12 students for each professor.

Table 2 indicates descriptive statistics for team success.

Table 2: Summary Statistics for Percent of Schools with Athletic Success per Year

	(1)			
	mean	sd	min	max
% of schools with $\geq$ 1 conference championship	35.24	5.71	23.69	42.14
% of schools with $\geq$ 1 NCAA tournament appearance	44.51	8.72	24.83	55.35
Observations	10			

Around 35% of schools have at least one of their 8 teams win a conference championship in any given year, while about 45% of schools have at least one of their 8 teams advance to the NCAA tournament, at the end of the year. This percentage is slightly higher because with at-large bids, non-conference winners may still participate in the end-of-season tournament. Though these averages seem high, this is indicative over eight different sports. Typically, for each sport, around 10% of teams are conference winners, and slightly more (say 12% are NCAA tournament participants. Assuming independence, the probability of a school having no teams win is $1-(0.9)^8 = 0.57$, and the probability of a school having no teams qualify is $1-(0.88)^8 = 0.64$. In reality, the independence assumption is not satisfied, as schools with better athletics programs are more likely to win, and even win multiple conference championships, reducing the number of available slots for other conference winners. Therefore, the actual percentages are much lower.

3.2 Treatment Assignment

In a differences-in-differences context, the treated group contains schools that qualified for the NCAA tournament, while the control group consists of schools that did not qualify for the NCAA tournament, in a given year. For any individual school, this treatment can switch on and off across years, where the school qualifies in one year, doesn't qualify the next, but qualifies again further down the road. Therefore, treatment is not permanent; once treated, a school is not always treated (unless they continue qualifying).

4 Methodology

4.1 Regression Model

The following models compare the effect of a school having at least one team qualify for the NCAA tournament on the response variable. The response variable is lagged two years to reflect athletics success in year t potentially influencing student decisions in year t + 1, and IPEDS student cohort data in year t + 2.

To estimate the effect of athletic success on yield rates, I started with the standard differences-in-differences model with two-way fixed effects:

$$Y_{i,t+2} = \beta_0 + \beta_1 NCAA_{it} + \gamma X_{it} + \alpha_i + \delta_t + \epsilon_{it}$$

where Y is the response variable $\in$ (Yield, Applicants), i is the school index, t is the year index, X_{it} is the vector of control variables representing school-level characteristics. β_1 measures the effect of participating in the NCAA tournament in the previous year. To incorporate fixed effects, we included α_i (school fixed effects) and δ_t (year fixed effects) to account for unobserved heterogeneity.

However, the standard DiD model is often biased when treatment effects are heterogeneous across units (schools) or over time. This is a major concern in this study, as the potential "Flutie Effect" may differ significantly across school types. In addition, the phenomenon is plausibly different between tournament appearances; theoretically, admissions shocks may be greater in magnitude following a school's first tournament appearance (at least in this time period) compared to a school's fifth tournament appearance. Due to the non-absorbing status of my treatment variable, differentiating between newly-treated and already-treated groups is essential.

Using the staggered DiD model that incorporates heterogeneous and intertemporal treatment effects introduced by Chaisemartin and D'Haultfœuille (2020), I also run regressions using this updated method to account for multiple treatment events and avoid using already-treated schools as potential controls.

5 Results

5.1 Pre-Trend Analysis

Before executing the regressions, I first ran an event-study to determine whether the parallel trends assumption was satisfied. In theory, yield rates and the number of applicants should be changing at similar rates prior to and immediately after a school has a team that qualifies for their respective NCAA tournament. I use the `did_multiplegt_dyn` estimator to estimate dynamic placebo coefficients.

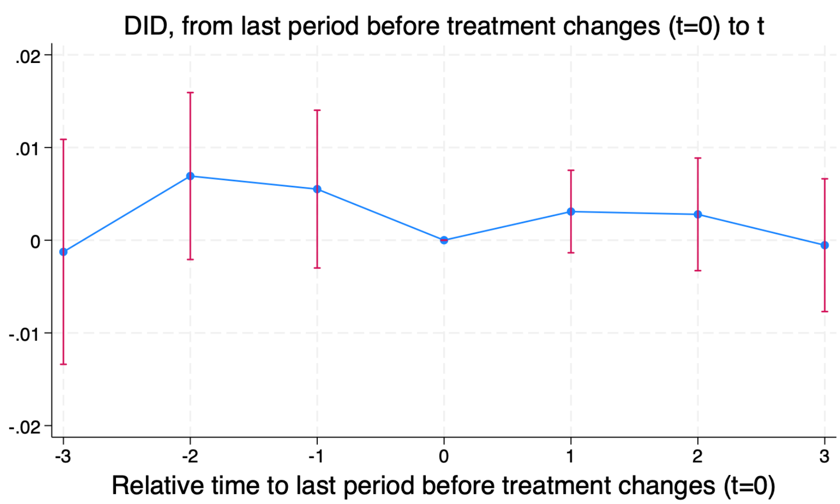

Figure 1: Event Study for Effect of NCAA Tournament Appearance on Yield Rate

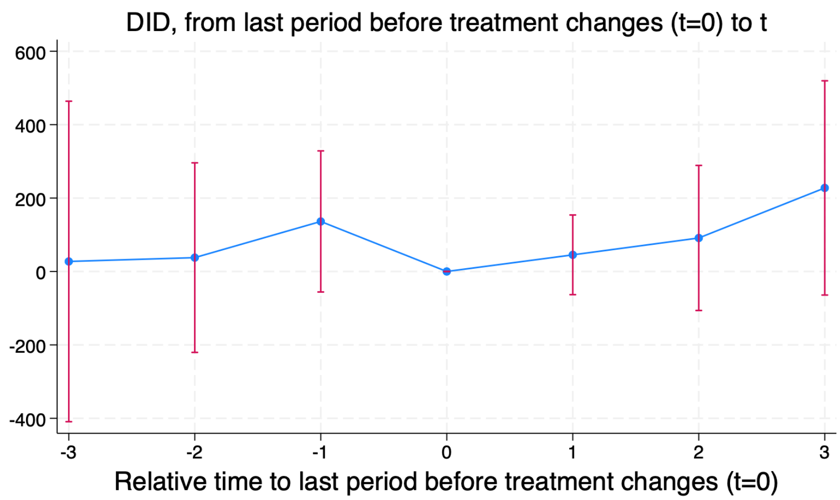

Figure 2: Event Study for Effect of NCAA Tournament Appearance on Applicants

The parallel trends assumption seems to be satisfied. The y-axes represent the difference in the outcome variable between treated and non-treated schools for a given point in time. In both event studies, the confidence interval at t = 0 contains 0, so we fail to reject the null hypothesis, and conclude that there is no statistically significant evidence of a difference in yield rate or applicants changes before and after the treatment is applied. The base year t = 0 is school-dependent; it is the last year before treatment switches for each school. Furthermore, the parallel trends line is relatively flat after 0, indicating that there is no substantive jump in the yield rate after the treatment shock. Interestingly, there seems to be a relatively large descent in yield rates and applicants the year right before the team qualified for the NCAA tournament. However, the Flutie effect in terms of applications does seem to come into play at t + 3 years, and the confidence interval almost does not contain 0.

5.2 Baseline DiD Regression

I then ran DiD regressions for each response variable. To illustrate differences in models, I compared the standard DiD model with Chaisemartin and D'Haultfœuille's.

Table 3: Effect of NCAA Tournament Qualification on Enrollment Outcomes

	TWFE		CDH	
	(1)	(2)	(3)	(4)
	Yield Rate	Log Applicants	Yield Rate	Log Applicants
NCAA Qualification (Lagged)	0.0004	−0.0019	0.0013	−0.0030
	(0.0022)	(0.0114)	(0.0030)	(0.0134)
Log Tuition (Adj.)	0.0074	0.1699		
	(0.0303)	(0.1797)		
Retention Rate	0.0013***	0.0014		
	(0.0003)	(0.0023)		
Student-Faculty Ratio	0.0006	0.0052		
	(0.0010)	(0.0117)		
Pct. Receiving Financial Aid	−0.0002	0.0003		
	(0.0003)	(0.0014)		
SAT 75th Percentile	−0.0001***	−0.0003		
	(0.0000)	(0.0002)		
Controls	Yes	Yes	Yes	Yes
School FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	2,790	2,790	2,304	2,304
R-squared	0.916	0.963		

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

I find no evidence that the coefficients are statistically significant. Having any of the 8 teams qualify for the NCAA tournament in year t - 2 is associated with a 0.04 percentage-point increase in the yield rate in year t. When using CH method, the magnitude slightly increases to a 0.13 percentage-point increase, but regardless, the economic impact is negligible. In terms of the number of applicants, I actually observe a negative association with NCAA qualification. Again, the CH method finds a slightly larger increase in magnitude, although the economic impact is again negligible. Having any of the 8 teams qualify for the NCAA tournament in year t - 2 is associated with a 0.30% decrease in applications in year t.

5.3 Regression by Sport

I then ran DiD regressions to examine the effect success in an individual sport has on yield.

Table 4: Regression of Yield Rate on Treatment by Sport (Canonical)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Yield	Yield	Yield	Yield	Yield	Yield	Yield	Yield
Baseball	-0.0035							
	(0.0032)							
MBB		-0.0021						
		(0.0024)						
MLax			-0.0014					
			(0.0047)					
MSoc				0.0056^{*}				
				(0.0030)				
Softball					-0.0077^{**}			
					(0.0036)			
WBB						-0.0018		
						(0.0032)		
WLax							0.0008	
							(0.0037)	
WSoc								0.0008
								(0.0046)
Observations	2790	2790	2790	2790	2790	2790	2790	2790
R-squared	0.9159	0.9159	0.9158	0.9160	0.9161	0.9158	0.9158	0.9158

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: CDH Estimates: Effect of NCAA Tournament Qualification on Yield Rate by Sport

	(1) Baseball	(2) MBB	(3) MLax	(4) MSoc	(5) Softball	(6) WBB	(7) WLax	(8) WSoc
NCAA	0.0000	0.0002	0.0003	−0.0000	0.0001	−0.0002	0.0003	0.0007
	(0.0026)	(0.0022)	(0.0055)	(0.0025)	(0.0024)	(0.0024)	(0.0048)	(0.0034)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,304	2,304	2,304	2,304	2,304	2,304	2,304	2,304

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Compared to the canonical DiD estimates, the CH estimates are much smaller in magnitude, and are practically 0, with a lack of statistical significance. However, when observing the canonical estimates, some sports have statistical significance, namely men's soccer and softball. In spite of this, qualifying for the NCAA tournament for any sport is associated with a change of no larger than 0.01 percentage points in the yield rate. This threshold is even lower with the CH method, with an associated change of no greater than 0.001 percentage points. Furthermore, several sports, namely baseball, men's basketball, men's lacrosse, men's soccer, and softball have differing signs between Tables 4 and 5, another indication of the biased nature of the standard DiD model.

I then ran DiD regressions to examine the effect success in an individual sport has on applications.

Table 6: Regression of Applicants on Treatment by Sport (Canonical)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Apps	Apps	Apps	Apps	Apps	Apps	Apps	Apps
Baseball	-0.0054							
	(0.0141)							
MBB		0.0166						
		(0.0133)						
MLax			-0.0066					
			(0.0222)					
MSoc				-0.0002				
				(0.0136)				
Softball					0.0009			
					(0.0170)			
WBB						-0.0014		
						(0.0133)		
WLax							-0.0066	
							(0.0185)	
WSoc								0.0148
								(0.0178)
Observations	2790	2790	2790	2790	2790	2790	2790	2790
R-squared	0.9628	0.9628	0.9628	0.9628	0.9628	0.9628	0.9628	0.9628

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: CDH Estimates: Effect of NCAA Tournament Qualification on Log Applicants by Sport

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Baseball	MBB	MLax	MSoc	Softball	WBB	WLax	WSoc
NCAA	0.0010	−0.0004	−0.0001	−0.0005	0.0001	0.0006	−0.0016	−0.0031
	(0.0102)	(0.0126)	(0.0170)	(0.0143)	(0.0110)	(0.0113)	(0.0240)	(0.0155)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,304	2,304	2,304	2,304	2,304	2,304	2,304	2,304

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Again, I observe that the magnitude of the coefficients from the CH method is much smaller than those for the canonical DiD method, along with a lack of statistical significance. In the CH method, the associated change in applications is no greater than 0.004% for any sport, while in the canonical method, this threshold is no greater than 0.02%. Also, baseball, men's basketball, women's lacrosse, and women's soccer have differing signs between Tables 6 and 7, indicating some biases involved.

So far, this robustness check corroborates the baseline findings of a negligible association with no evidence of statistical significance between qualifying for the NCAA tournament and admissions.

5.4 Regression by Gender

I additionally run DiD regressions comparing men's vs. women's sports, starting with the standard method.

Table 8: Regression of Treatment by Gender (Canonical)

	Men		Women	
	(1)	(2)	(3)	(4)
	Yield	Log Applicants	Yield	Log Applicants
Men	0.0015	0.0107		
	(0.0023)	(0.0107)		
Women			-0.0023	-0.0019
			(0.0025)	(0.0121)
Observations	2790	2790	2790	2790
Adj. R^2	0.9016	0.9565	0.9016	0.9564

Table 9: CDH Estimates: Effect of NCAA Tournament Qualification on Admissions by Gender

	(1)	(2)	(3)	(4)
	Yield	Log Applicants	Yield	Log Applicants
	Men		Women	
NCAA	0.0002	−0.0003	0.0014	−0.0043
	(0.0026)	(0.0111)	(0.0028)	(0.0141)
Controls	Yes	Yes	Yes	Yes
School FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	2,304	2,304	2,304	2,304

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Corroborating previous results, we observe a coefficient near 0 for the primary treatment variables, without evidence of statistical significance. Independent of gender, teams that succeed athletically by qualifying for the NCAA tournament are associated with incremental increases in the yield rate and similar decreases in the number of applicants. The associated change in yield rate does not exceed 0.01 percentage points, while the associated change in applicants does not exceed 1%. It is worth noting, however, that the canonical DiD regression has a substantially larger (35x) coefficient for men's sports, where a men's team qualifying for the NCAA tournament is associated with a 1% increase in applications for the following year. However, as mentioned previously, the canonical method is often biased due to heterogeneous treatment effects, and the CH regression indicates that the applicants coefficient is more in line with previous estimates, with an associated change closer to 0. All in all, the similarity in magnitude and direction across the two robustness checks, differentiating by sport and gender, indicates that the baseline findings are not being driven by any particular sport or by gender. Therefore, effects appear to be uniform across the 8 sports described and men's vs. women's sports.

5.5 Regression by Sport x Expenses

I am interested in determining whether the effect on admissions is amplified by greater spending for any given sport, starting with the yield rate. Because the CH method only accepts a binary treatment variable, I am unable to utilize interaction terms for that method.

Table 10: NCAA × Expenses on Yield Rate

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Yield	Yield	Yield	Yield	Yield	Yield	Yield	Yield
Base*Exp	-0.00713							
	(0.00720)							
MBB*Exp		-0.00668						
		(0.00547)						
MLax*Exp			0.00278					
			(0.0103)					
MSoc*Exp				0.00691				
				(0.00698)				
Soft*Exp					-0.0107			
					(0.00695)			
WBB*Exp						-0.00255		
						(0.00846)		
WLax*Exp							0.00123	
							(0.00863)	
WSoc*Exp								-0.00373
								(0.0111)
N	2302	2415	1393	2409	2356	2439	1642	2423

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

None of these interaction terms are statistically significant, and the magnitudes do not exceed a 0.02 percentage point change. Baseball, men's basketball, softball, women's basketball, and softball have negative coefficients, indicating a 1% increase in expenses is associated with a slightly smaller effect on yield. Conversely, men's and women's lacrosse, and men's soccer have positive coefficients, indicating a 1% in expenses is associated with a slightly larger effect on yield. From table 5, these sports also have a positive association with yield, suggesting that further investment in these sports may be associated with larger increases in yield. However, given that none of these estimates are statistically significant, no definitive conclusions can be drawn.

I then determine whether the effect on applicants is also amplified by sport-specific spending.

Table 11: NCAA × Expenses on Applicants

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Apps	Apps	Apps	Apps	Apps	Apps	Apps	Apps
Base*Exp	0.0272							
	(0.0263)							
MBB*Exp		0.0126						
		(0.0335)						
MLax*Exp			-0.0642					
			(0.0429)					
MSoc*Exp				0.0518**				
				(0.0239)				
Soft*Exp					0.00567			
					(0.0337)			
WBB*Exp						0.0258		
						(0.0315)		
WLax*Exp							0.108***	
							(0.0399)	
WSoc*Exp								0.0266
								(0.0443)
N	2302	2415	1393	2409	2356	2439	1642	2423

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

With the exception of men's lacrosse, all other sports have positive coefficients, indicating that a 1% increase in expenses is associated with a slightly larger effect on applicants. In addition, men's soccer and women's lacrosse are statistically significant, although due to the biased nature and flaws of the canonical DiD model, I am unsure whether this is actually true. From table 7, baseball, softball, and women's basketball also have positive associations with applicants, suggesting that increased spending for those sports may be associated with larger increases in applications. Again however, due to a lack of statistical significance, no definitive conclusions can be drawn.

6 Discussion

Despite additional robustness checks by differentiating schools by sport, and then by gender, we observe an overall lack of statistical significance in determining whether there is an association between athletics success and admissions for Division III schools. The yield rate coefficient tends to be slightly positive (increase), although the coefficient for the number of applicants tends to be slightly negative (decrease). Therefore, fewer students wish to apply to the school, but conditional on gaining admission, more students wish to enroll. This paradox is fascinating, as one would think the two outcome variables would both have the same correlation with athletics success. It is possible athletics success may be a differentiator when students must choose which college to attend, though the effects may be less important when it comes to applying. Given the majority of research at the Division I level, this analysis adds an additional lens to current research on the relationships between collegiate athletics and admissions at the Division III level.

There are many limitations with this paper. First, data from the NCAA only dates back to 2013 for the sports I chose, including automatic qualifiers and at-large bids. However, there is overall data regarding which teams qualified for the NCAA tournament, which dates further back. Additional years of data would be helpful, as I only have 9 (minus the Covid year in 2020). In addition, when comparing tournament qualification data with the list of current D3 teams, I realized that some teams that qualified were not on the D3 list. This was due to a variety of reasons, such as transitioning to a different division level or the school closing down. Though these events occurred within my time period, I opted to omit these schools from my study in order to maintain a balanced panel dataset for all schools in all years. Finally, one of the key assumptions with the Differences-in-Differences technique is the pre-trends assumption, and given the short time-frame allotted, there are concerns over whether this prerequisite is satisfied to proceed with analysis.

Furthermore, there are also issues with endogeneity and the structure of the study. It is difficult to establish causality because of the issue of reverse causality; schools with a more selective yield, or more applicants, are plausibly better funded, have more resources, and can build a better athletic program, thus leading to more NCAA tournament appearances. In addition, it is difficult to account for all possible variables a high school student considers when deciding whether to apply to a school, and whether to attend if admitted. Thus, there is omitted variable bias due to unobserved school-level characteristics that are not included as control variables in my study. Finally, treatment assignment is not random; it is reasonable to assume that tournament participants systematically differ in quality and other characteristics compared to non-tournament participants. Finally, I was unable to find data for an exogenous instrument or threshold that could plausibly affect athletics success but not admissions, as an instrumental variables or regression discontinuity approach would also help address these endogeneity concerns.

There are also several additional robustness checks I could conduct to verify these initial findings. Firstly, I could conduct regressions based on the quantity of NCAA tournament appearances each school has per year. This would determine whether the effect on yield and applicants is different for schools that have multiple teams qualify for the NCAA tournament. Moreover, as mentioned previously, I decided to lag the athletics succeess variable by two years, though effects may vary if the lagged dependent variable is by more years. However, I believe the potential effect would be short-term in nature. In addition, there is a concern that schools that continually qualify for the NCAA tournament are benefitting from a weak conference, rather than a strong team. Team A may be better than Team B, but if Team A plays in a strong conference, doesn't win, and does not gain an at-large bid, they will not qualify, while Team B is weaker, but by virtue of their conference competition being lower, gain an automatic bid and qualify. Finally, schools differ in name brand and recognition, something that is hard to quantify. It is certainly plausible that students choose to apply/attend based on their perception of an institution's academic reputation, rather than the success their athletic programs have on the field.

Summing up, it is important to note the differences between the canonical DiD method and the staggered DiD method introduced by Chaisemartin and D'Haultfœuille. Though there were some statistically significant terms in the standard DiD setup, after accounting for multiple treatment periods, heterogeneity within schools and sports, and other biases, I find an overall lack of statistical significance and negligible admissions outcomes associated with collegiate athletics success at the Division III level. Further robustness checks with sport-specific spending indicate that certain teams may amplify a positive association between NCAA tournament qualification and admissions, although due to a lack of statistical significance, it is unclear whether institutions can adjust their policies.

7 Works Cited

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